Q1:NOV 2011 P2

9 **(a)** Define

(i) *half life*,

____

____

(ii) *activity* of a radioactive isotope.

____

______________________________________ [2]

(b) A radioactive material has an activity of 6×10^5 Bq and 9.0×10^{15} undecayed particles.

Calculate

(i) the decay constant, λ.

$\lambda =$ ____________________

(ii) the half life, $t\frac{1}{2}$.

$t\frac{1}{2} =$ ____________________ [4]

Q2:NOV 2004 P2

8 **(c)** In a radioactivity experiment, a sample contains 3 g of a radioisotope which has a half-life of 42 seconds. Determine the mass of the radioisotope which remains after 60 seconds.

mass = ______________________________ [2]

Q3:JUN 2003 P2

6 **(a)** A student is provided with a freshly prepared sample of a radioactive material and the count rate *C* from the source is found to vary with time *t* as shown in Fig. 6.1(a).

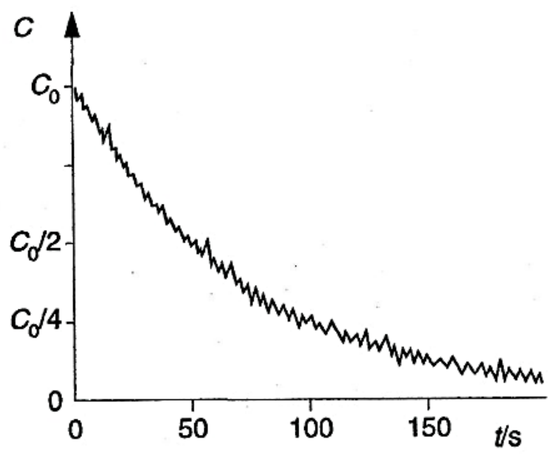

Fig. 6.1(a)

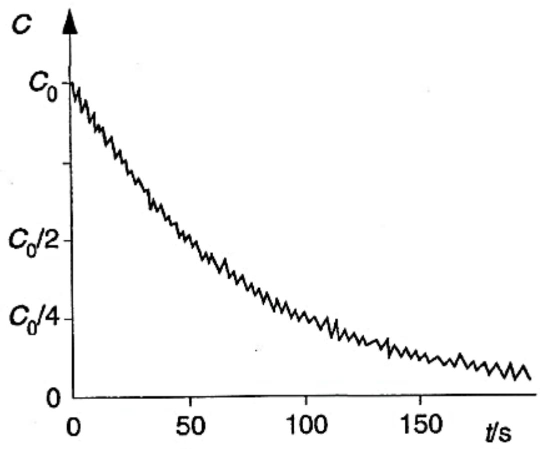

Fig. 6.1(b)

A second similar sample of the radioactive material is then prepared and the student repeats the experiment, but with the sample at a higher temperature. The variation with time of the count rate for the second sample is shown in Fig. 6.1(b).

State the evidence that is provided by these two experiments for

(i) the random nature of radioactive decay,

....

....

(ii) the spontaneous nature of radioactive decay.

....

....
[2]

(b) The radioactive source in **(a)** is an isotope of radon ($^{220}_{86}Rn$) that emits α-radiation to become polonium (Po).

(i) State the number of neutrons in one nucleus of radon-220.

number = ... [1]

(ii) Write down a nuclear equation to represent the radioactive decay of a nucleus of radon.

$$^{220}_{86}Rn \rightarrow \,^{\cdots\cdots}_{\cdots\cdots}Po + \,^{\cdots\cdots}_{\cdots\cdots}\cdots\cdots$$

[3]

Q4:JUN 2016 P2

9 (a) The decay of a radioactive isotope is said to be *random* and *spontaneous*.

Explain what is meant by a *spontaneous process*.

____

____

____

____ [1]

(b) Thorium-231, $^{231}_{90}\text{Th}$ decays to form Protactinum-231, $^{231}_{91}\text{Pa}$.

(i) Write down the nuclear equation representing this decay.

____

(ii) During this decay, a β- particle and an X-ray photon are emitted with energies of 5.32 MeV and 0.50 MeV respectively.

Calculate the

1. mass equivalence of the energy released during the decay,

mass = ____________

2. wavelength of the emitted X-ray photon.

wavelength = ____________

[5]

Q5:NOV 2013 P2

9 **(a)** Define the following terms:

(i) *half life*

____

____

____

(ii) *activity*

____

____

____ [2]

(b) Cobalt-60 has a half life of 5.5 years.

Determine the activity of

(i) 0.002 g of cobalt-60,

activity = ______________________

(ii) the cobalt-60 in (i) four years later.

activity = ______________________ [4]

Q6:JUN 2011 P2

5 (a) In an α-scattering experiment an α-particle was directed towards the centre of a gold, $\left({}^{197}_{79}\text{Au}\right)$, nucleus as shown in **Fig. 5.1**. The α-particle moves with a kinetic energy of 4.8 MeV towards the nucleus.

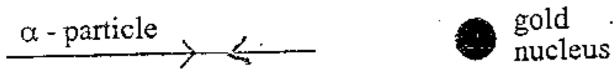

Fig. 5.1

(i) Sketch on **Fig. 5.1** the path of the α-particle.

(ii) Determine the distance of closest approach of the α-particle to the gold nucleus.

distance of closest approach = ______________ [4]

(b) State **three** uses of radioisotopes.

1. ______________________________

____

2. ______________________________

____

3. ______________________________

______________________________ [3]

Q7:NOV 2009 P2

9 (a) Give **three** uses of radio isotopes.

1. ______________________________

2. ______________________________

3. ______________________________ [3]

(b) A certain isotope decays to $\frac{1}{3}$ of its original value in 10 days.

Determine

(i) its decay constant,

decay constant = ______________________

(ii) its half life.

$T_{\frac{1}{2}}$ = ________________ [4]

Q8:NOV 2018 P2

9 The radioactive decay of an isotope Thorium (Th) is represented by the equation,

$$^{225}_{90}TH \rightarrow {}^{A}_{Z}Ra + \alpha$$

The Thorium isotope has a half-life of 8.0 minutes.

(a) Define the term **half-life**.

____

____ [1]

(b) (i) State what the symbols **A** and **Z** in the equation represent.

A ______________________________

Z ______________________________

(ii) Determine the numerical values of,

A ______________________________

Z ______________________________

(iii) Calculate the decay constant of the Thorium isotope.

[6]

(c) Give any **three** uses of radioisotopes.

____

____

____

______________________________ [3]

Q9:JUN 2018 P2

8 The half life of a sample of radio active isotope is 8 hours. A freshly prepared sample of this isotope contains 10^{15} atoms.

Calculate the

(a) initial activity,

(b) number of radioactive atoms remaining after 2 hours.

number of atoms ______________________________ [2]

(c) State any **two** uses of radioactivity.

1.____

2.____ [2]

Q10:JUN 2017 P2

8 **(a)** Radioactive decay is said to be spontaneous and random.

Explain the underlined terms.

1. spontaneous ______________________________________

____

2. random______________________________________

______________________________________ [2]

(b) A sample of cobalt-60 has 3.72×10^{21} particles. Cobalt-60 has a decay constant of $1.83 \times 10^{-9} s^{-1}$.

Calculate the

(i) mass of cobalt-60 present,

mass = ______________

(ii) half life, $t_{\frac{1}{2}}$, of the source,

$t_{\frac{1}{2}}$ = ______________

(iii) number, N, of undecayed particles after 1 year.

N = ______________

[6]

(c) Explain why high energy α-particles are not suitable for bombarding atoms to cause nuclear reactions.

____

____

____

____ [2]

Q11:NOV 2015 P2

9 (a) Radioactive decay is *random* and *spontaneous.*

Define the terms in italics.

spontaneous ____

____

random ____

____ [2]

(b) Radium-226 has a decay constant of 1.8×10^{-6} s and an initial decay rate of $5.2 \times 10^{10}\ s^{-1}$

Calculate the

(i) initial number of radioactive atoms present,

number of atoms = ____________________

(ii) time taken for the activity to fall to a quarter of its initial value.

time = ____________________

[4]

Q12:JUN 2014 P5

4 **Table 4.1** gives some of the properties of different radiations.

Table 4.1

radiation	penetrating ability	effect of magnetic field
alpha (α)		deflection in the same direction as positive charge
gamma ray (γ)	several cm of dense metal	
beta (β)	few mm of aluminium	

(a) Complete **Table 4.1**. [3]

(b) Give **two** sources of Background radiation. [2]

(c) Describe how the penetrating ability of beta particles can be used to monitor the thickness of aluminium sheets uniform during manufacture. [3]

(d) The half-life of ^{90}Sr is 28 years. When its activity falls to 15% of its original value it should be replaced.

(i) Determine how many years later should it be replaced.

(ii) Suggest **one** safe way of disposing of ^{90}Sr after use.

[4]

Q13:JUN 2009 P5

1 (c) (i) 'A beta emitting radioactive source has a half-life of 13 hours.'

1. Explain this statement.

2. Determine the percentage loss of activity which occurs in the source over a period of 2.0 hours.

(ii) Describe how a radioactive source can be used in the manufacture of thin uniform sheets of aluminium.

[8]

Q14:NOV 2008 P5

5 (a) Define the terms *half-life* and *activity*. [2]

(b) A 10.0 kg sample from a living plant has an activity of 30 000 counts per second due to carbon-14. A 500 g sample of the same type of plant but dead has an activity of 1 000 counts per second.

(i) Explain why carbon-14 is used in dating.

(ii) Calculate the age of the dead sample, given that the half-life of carbon-14 is 5 568 years.

(iii) Explain why the value calculated gives the time after the plant is dead. [7]

(c) The plant in **(b)** is believed to have lived for 2 000 years.

Sketch the graph of activity against time from the time the plant was alive to date. [3]

Q15:NOV 2015 P5

5 (a) (i) State any **two** sources of background radioactive radiation.

(ii) **Table 5.1** shows properties of alpha and beta radiations.

Copy and complete **Table 5.1.**

Table 5.1

radiation	penetration	ionising ability	effects of electric field
alpha (α)			deflected slightly in the same direction as +ve charge.
beta (β)	many cm of air at s.t.p. few mm of light metals such as aluminium		strong deflection in opposite direction to α.

(iii) Explain why gamma rays are **not** used in paper thickness control during manufacturing. [7]

(b) (i) Iodine-131 has a half-life of 8 days.

Calculate the number of iodine-131 atoms remaining after 24 days for a sample with 1.0×10^{10} atoms. [5]

(ii) Explain how radiation can be harmful to a human being.

Q16:JUN 2015 P5

5 (a) Radioactive decay is a random and spontaneous process.

Explain what is meant by a

(i) *random process,*

(ii) *spontaneous process.*

[2]

(b) One isotope of lead, $^{214}_{82}Pb$, has a half-life of 27 minutes.

(i) Define an *isotope,*

(ii) Determine the constituents of $^{214}_{82}Pb$ nucleus from its mass and atomic numbers.

[3]

Q17:NOV 2014 P5

1 (b) (i) Two radioisotopes, A and B, were mixed and the activity of the mixture was measured over a period of time, *t*. **Fig. 1.1** shows the natural logarithm of the activity plotted against time.

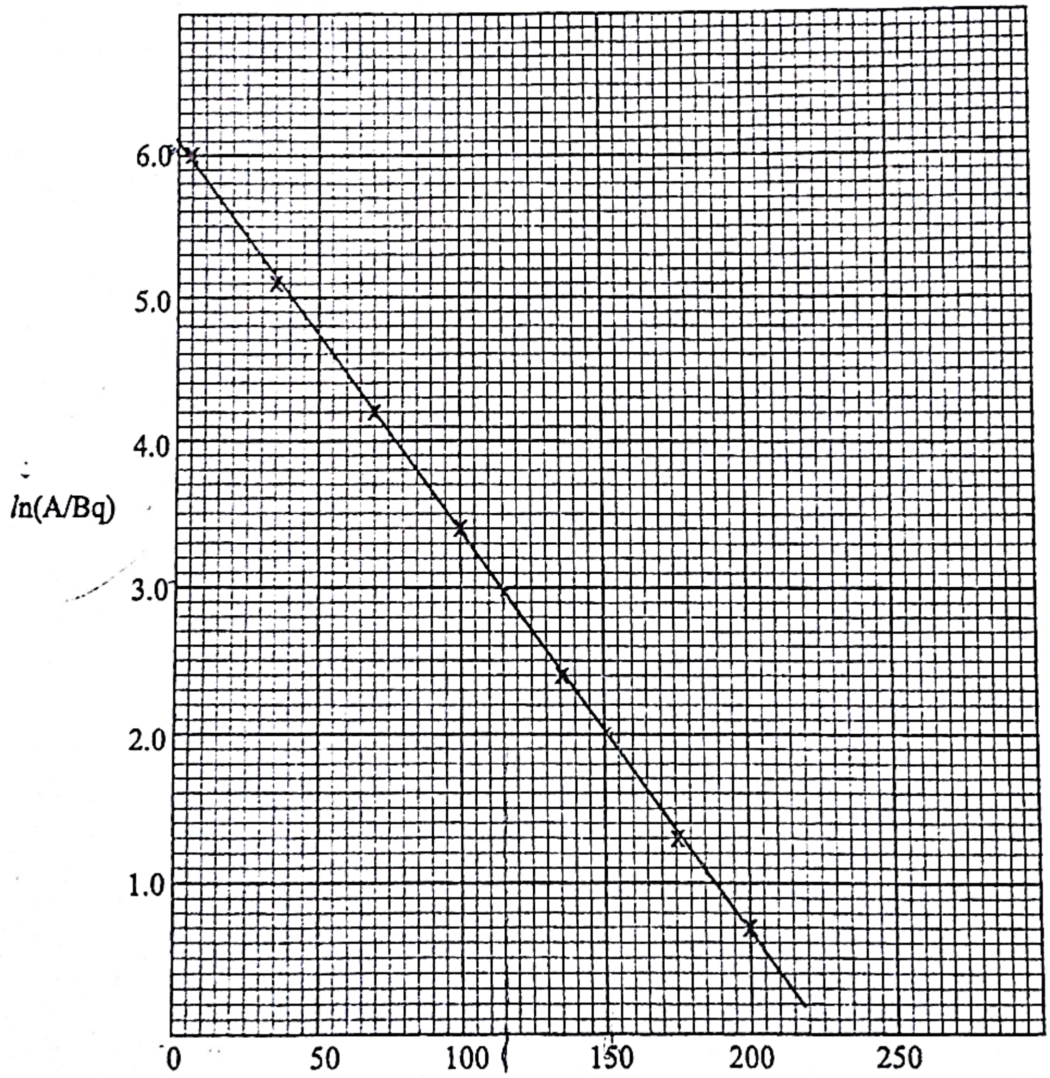

Fig. 1.1

1. Explain why the gradient of the graph is negative.
2. Determine the decay constant of the mixture.

(ii) Radioisotopes A and B decay by α-emission. The energies of the α-particles from A and B were found to be 5.2 MeV and 9.3 MeV respectively.

Giving reasons, state which of the isotopes A or B has a shorter half-life.

[6]

Q18:JUN 2013 P5

1 (e) (i) Define the term *decay constant.*

(ii) The half-life of radon-220 is 55.7 s and there are 6.0 g of radon-220 initially.

Calculate the number of radon-220 atoms remaining after three and half days.

(iii) State the effects of using, in a hospital, a short half-life source and a long half-life source.

[5]

Q19:NOV 2012 P5

1 (a) (i) Define the decay constant, λ.

(ii) Sketch a graph to show how the activity of a radioactive isotope of half-life 30 days varies with time from an initial value A.

(iii) A lubricating oil contains a radioactive isotope X of half-life 25 days and another isotope Y of half-life 5 hours. Initially the oil has an activity of 16 kBq. After 40 days the activity of the oil falls to 2 kBq.

Estimate the initial percentage activity of isotope Y in the mixture.

[8]

Q20:JUN 2018 P5

1 (d) Iodine-131, has a decay constant of $1.003 \times 10^{-6} s^{-1}$.

(i) Define the term *decay constant.*

(ii) A patient was given 131 mg of the iodine drug.

Calculate the activity of the drug at the time of ingestion.

(iii) Justify the use of Iodine-131 as a tracer to diagnose the thyroid gland disorders.

[6]

Q21:NOV 2002 P3

6 (a) In terms of the constituents of atomic nuclei, explain the meaning of

(i) atomic number,

(ii) mass number,

(iii) isotopes. [3]

(b) Account for the fact that, although nuclei do not contain electrons, some radioactive nuclei emit beta particles. [2]

(c) Cobalt has only one stable isotope ^{59}Co. What form of radioactive decay would you expect the isotope ^{60}Co to undergo to form ^{59}Co? Give a reason for your answer. [2]

(d) ^{60}Co decays to ^{59}Co with a half-life of 5.3 years.

(i) What is the activity of a source containing 0.015 g of ^{60}Co?

(ii) What is the activity of the source 3 years later? [